CSCI 5526

Risk Stratification

Announcements

- MP2 due on 10/9!
- Self-sign up for final project groups on Canvas!
  - Due on 10/16/26!

Student Questions

Is the elbow method frequently used in practice? It seems like if cross validation is possible, then that would be better. Is the main drawback to cross validation cost?

Yes, elbow method is used with algorithms like k-means.

Cross-validation approach is more robust because we are testing clustering quality on unseen data. But it requires performing clustering many times.

Student Questions

In the Zeisel et al. study (P14), if the clustering itself used the full dataset, could information from the held-out cells indirectly enter the cluster labels? Does this mean the random forest mainly measures cluster robustness or separability rather than true generalization to new data? Can you talk more about potential data leakage issues in ML?

-- Yes, that is a good observation.

-- Authors did clustering on the whole data and then evaluated the robustness using RF.

-- It measures cluster stability and not necessarily generalization.

Outline

- Evaluating clustering results
- Risk Stratification

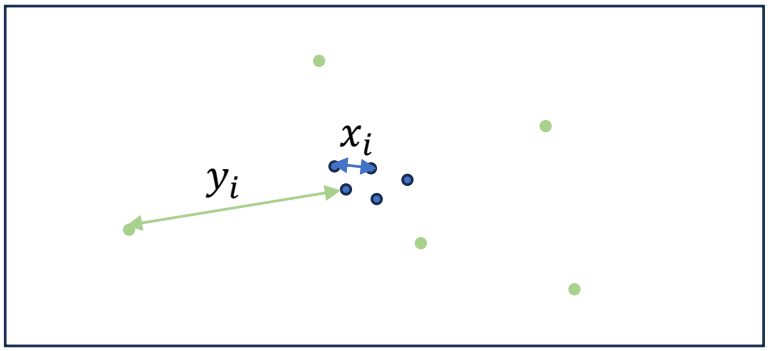
Evaluating Clustering

Slides adapted from Jiawei Han, Micheline Kamber, and Jian Pei

Assessing Clustering Tendency

- Assess if non-random structure exists in the data by measuring the probability that the data is generated by a uniform data distribution
- Test spatial randomness by statistic test: Hopkins Static
  - The Hopkins Statistic:

$$H = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i}$$



  - If there is no structure, H is close to 0.5. If there is structure, H is close to 1.

  - In Python: https://pypi.org/project/pyclustertend/

Determining the Number of Clusters

- Elbow method
  - Use the turning point in the curve of sum of within cluster variance with respect to the # of clusters (k).
- Cross validation method
  - Divide a given data set into m parts.
  - Use m-1 parts to obtain a clustering model.
  - Use the remaining part to test the quality of the clustering.
    - E.g., for each point in the test set, find the closest centroid, and use the sum of squared distance between all points in the test set and the closest centroids to measure how well the model fits the test set.
  - Repeat it m times, compare the overall quality measure with respect to different k's, and find # of clusters that fits the data the best.

Measuring Clustering Quality

- Two methods: extrinsic and intrinsic
- Extrinsic: supervised, i.e., the ground truth is available.
  - Compare a clustering against the ground truth using certain clustering quality measure.
- Intrinsic: unsupervised, i.e., the ground truth is unavailable.
  - Evaluate the goodness of a clustering by considering how well the clusters are separated, and how compact the clusters are.

Extrinsic Methods

- Clustering quality measure: Q(C, Cg), for a clustering C given the ground truth Cg.

- Q is good if it satisfies the following criteria.
  - Cluster homogeneity: A clustering result satisfies homogeneity if all of its clusters contain data points which are members of a single class only.

  - Cluster completeness: A clustering result satisfies completeness if all the data points that are members of a given class are elements of the same cluster.

- v-measure (beta is a weight specified in [0,1])

```
v = (1 + beta) * homogeneity * completeness / (beta * homogeneity + completeness)
```

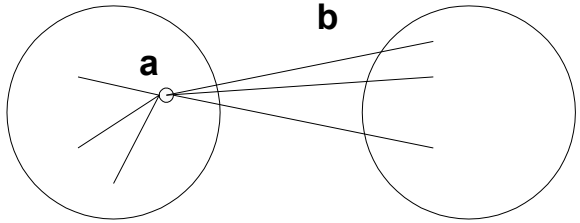
Intrinsic Measures: Silhouette Coefficient

- Silhouette Coefficient: a measure based on intra-cluster distances and inter-cluster distances.

- For an individual point, i
  - Calculate a = average distance of i to the points in its cluster
  - Calculate b = min (average distance of i to points in another cluster)
  - The silhouette coefficient for a point is then given by

$$s = \frac{b - a}{b}$$

  - Typically, between 0 and 1.
  - The closer to 1 the better.



- Can calculate the average Silhouette width for a cluster or all clusters.

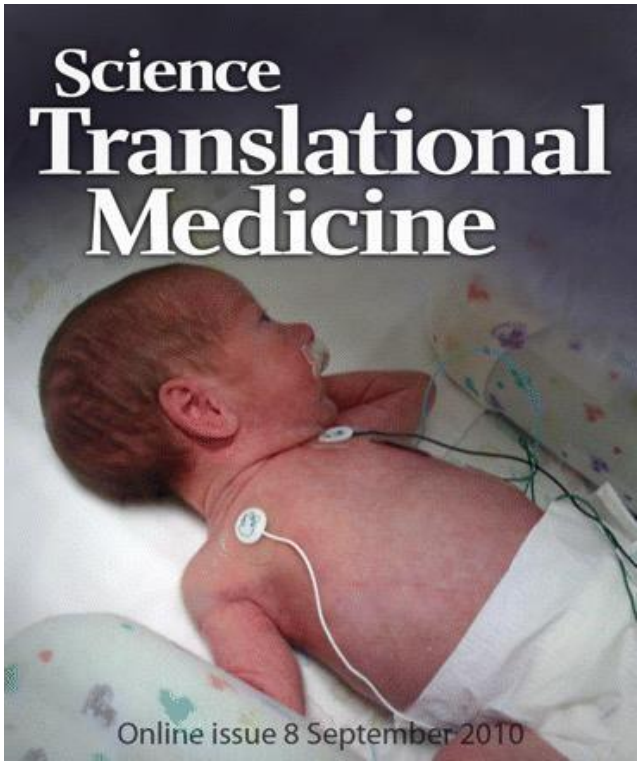
Risk Stratification

(Slides from David Sontag, MIT)

What is Risk Stratification?

- Separate a patient population into high-risk and low-risk of having an outcome
  - Predicting something in the future
- Coupled with interventions that target high-risk patients
- Goal is typically to reduce cost and improve patient outcomes

Examples



- Preterm infant's risk of severe morbidity?
- This prediction algorithm was developed with electronically captured physiological time series data from the first 3 hours of life in preterm infants.

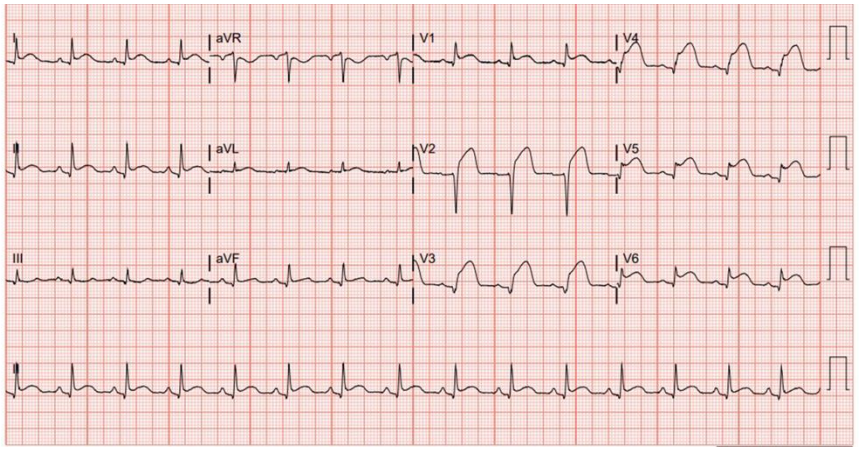
Saria et al., Science Translational Medicine 2010

Examples



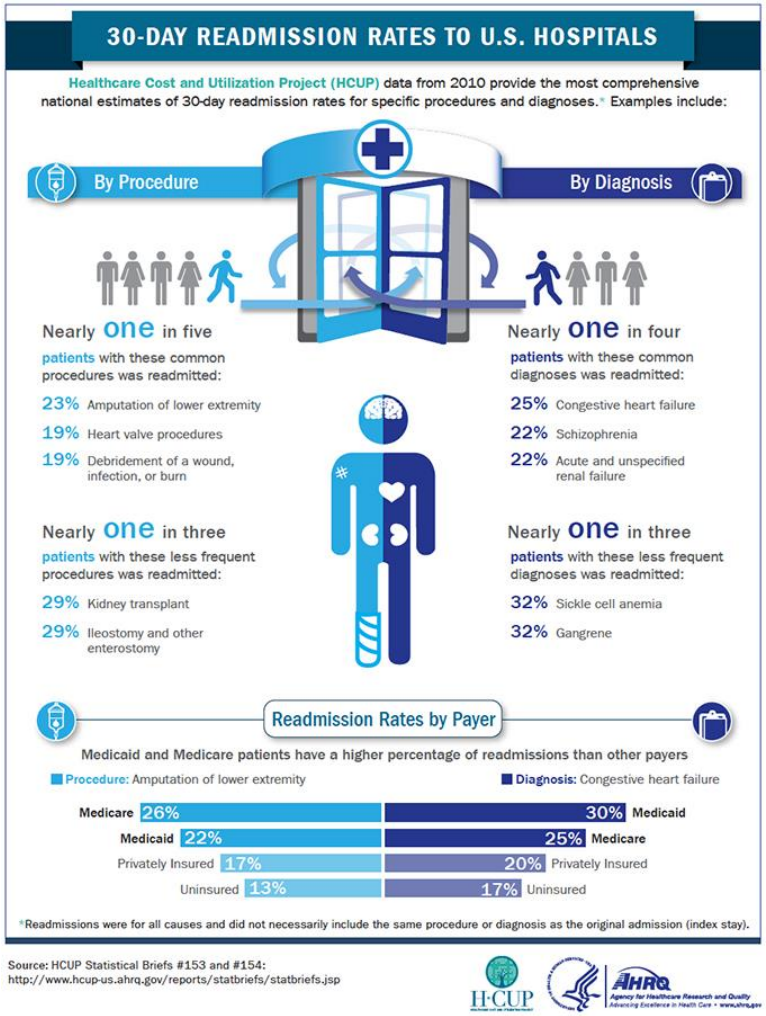
Source: https://www.drmani.com/heart-attack/ (top)

- Does this patient need to be admitted to the ICU?



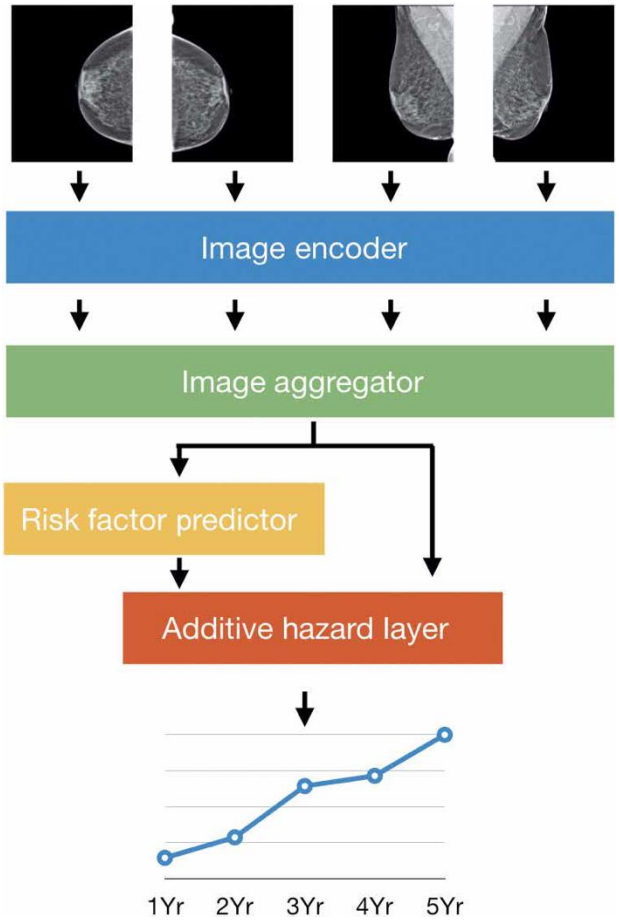
Pozen et al., NEJM 1984

Examples



Likelihood of hospital readmission?

Examples



- Will this woman develop breast cancer in the next 5 years?

Yala et al., Science Translational Medicine 2021

Examples



David, your genetics are associated with a **typical likelihood** of developing type 2 diabetes.

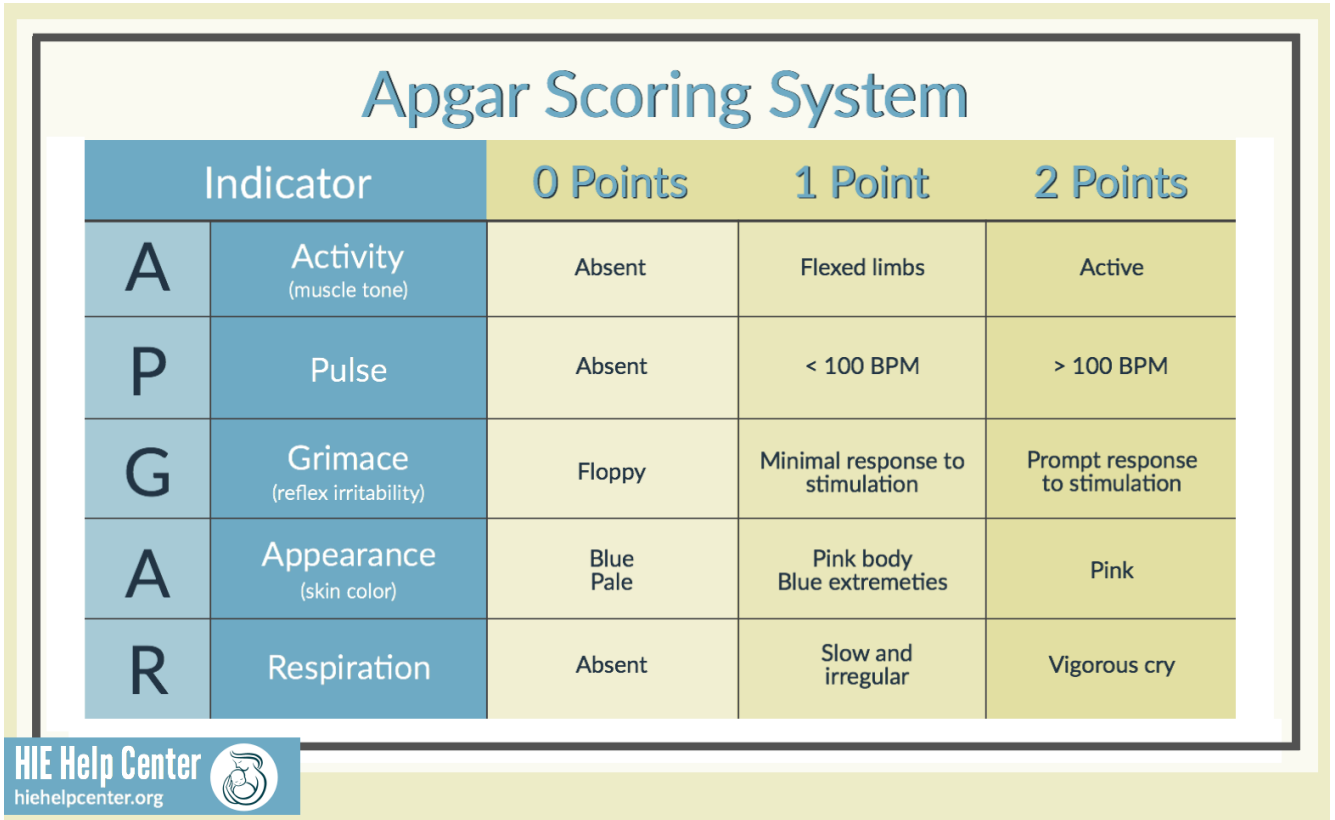
Based on data from 23andMe research participants, people of European descent with genetics like yours have an estimated **22% chance** of developing type 2 diabetes at some point **between the ages of 37 (your current age) and 80.**



Differential Diagnosis vs Risk Stratification

Differential Diagnosis	Risk Stratification
Often iterative/active	Often passive
Considers a large number of possible conditions	One condition

Old Approach



Apgar Scoring System

Indicator		0 Points	1 Point	2 Points
A	Activity (muscle tone)	Absent	Flexed limbs	Active
P	Pulse	Absent	< 100 BPM	> 100 BPM
G	Grimace (reflex irritability)	Floppy	Minimal response to stimulation	Prompt response to stimulation
A	Appearance (skin color)	Blue Pale	Pink body Blue extremeties	Pink
R	Respiration	Absent	Slow and irregular	Vigorous cry

HIE Help Center
hiehelpcenter.org

Old Approach

PATIENT HEALTH QUESTIONNAIRE (PHQ-9)

NAME: _____ DATE: _____

Over the last 2 weeks, how often have you been bothered by any of the following problems? *(use "✓" to indicate your answer)*	Not at all	Several days	More than half the days	Nearly everyday
1. Little interest or pleasure in doing things	0	1	2	3
2. Feeling down, depressed, irritable, or hopeless	0	1	2	3
3. Trouble falling or staying asleep, or sleeping too much	0	1	2	3
4. Feeling tired or having little energy	0	1	2	3
5. Poor appetite, weight loss, or overeating	0	1	2	3
6. Feeling bad about yourself - or that you are a failure or have let yourself or your family down	0	1	2	3
7. Trouble concentrating on things, such as school work, reading, or watching television	0	1	2	3
8. Moving or speaking so slowly that other people could have noticed. Or the opposite - being so fidgety or restless that you have been moving around a lot more than usual	0	1	2	3
9. Thoughts that you would be better off dead, or of hurting yourself in some way	0	1	2	3
add columns:		+	+	
TOTAL:				

Score	Depression Severity
0-4	Minimal
5-9	Mild
10-14	Moderate
15-19	Moderately severe
20-27	Severe

Old Approach vs New Approach

- Traditionally, risk stratification was based on simple scores using human-entered data.

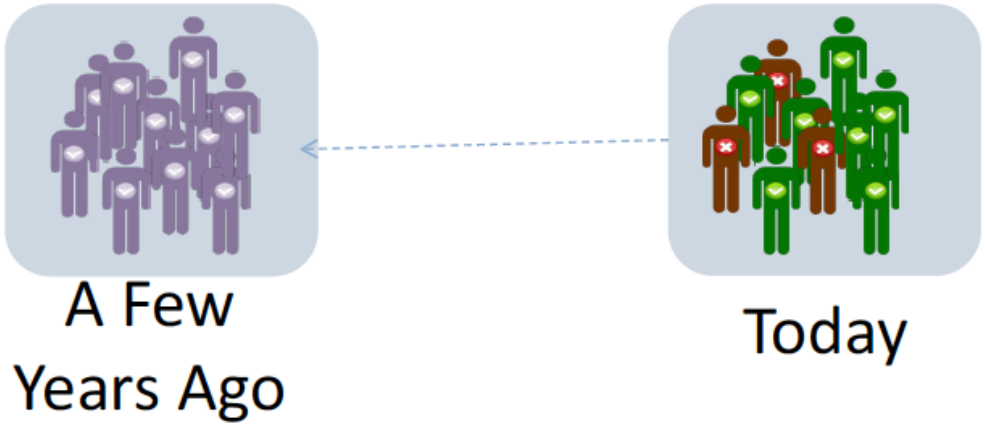
- Now, based on machine learning on high-dimensional data
  - Fits more easily into workflow
  - Higher accuracy
  - Quicker to derive

New Approach: What is Needed?

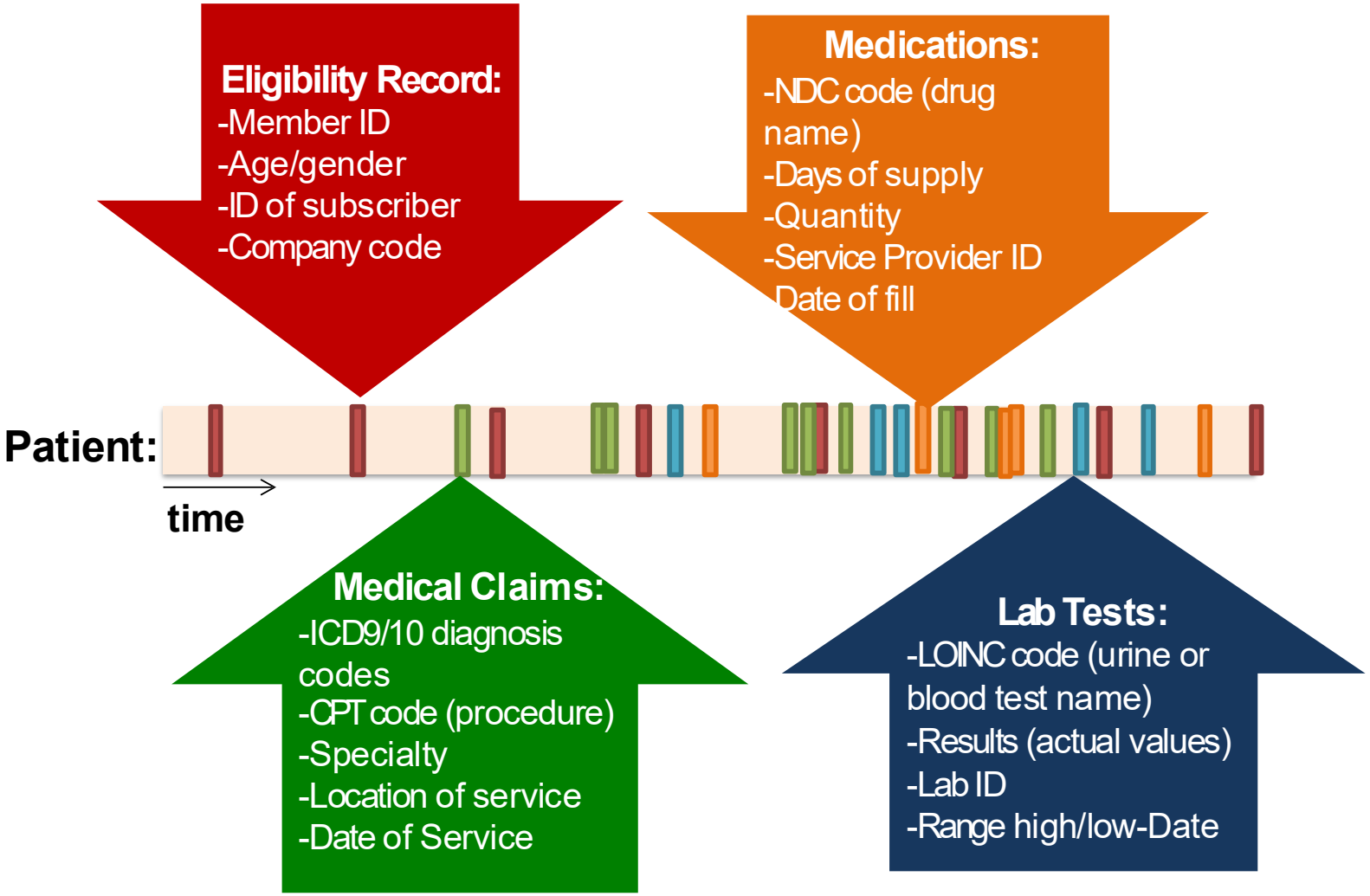
- Specification of prediction time/index date
- A way of encoding the data we have on the patient
  - CNN for images
  - Bag of words for text document
  - Longitudinal structured data…?
- A target, typically derived from the EHR
- Choice of appropriate supervised ML algorithm
  - Regression? Classification?

How to Model?

- Consider predicting the risk of diabetes.
- Looking at individuals who got diabetes today, (compared to those who didn't)
  - Can we infer which variables in their record could have predicted their health outcome?



Administrative & Clinical Data



Claims Data Characteristics

- Sparse data
  - Vast coding spaces for diagnoses, symptoms, procedures, medications, labs
  - Most patients don't have most of these
- Visit-level temporality
  - Data collected only at interactions with the health care system; highly variable intervals
- Long-term dependencies
  - How to encode these? LSTM, bi-RNN, CNN, attention, …
  - simpler trends

Kodialam RS, Boiarsky R, Sontag D. Deep contextual clinical prediction with reverse distillation. AAAI. 2021; Available from: http://arxiv.org/abs/2007.05611v1

Top diagnosis codes

Disease	count
401.1 Benign hypertension	447017
272.4 Hyperlipidemia NEC/NOS	382030
401.9 Hypertension NOS	372477
250.00 DMII wo cmp nt st uncntr	339522
272.0 Pure hypercholesterolem	232671
272.2 Mixed hyperlipidemia	180015
V72.31 Routine gyn examination	178709
244.9 Hypothyroidism NOS	169829
780.79 Malaise and fatigue NEC	149797
V04.81 Vaccin for influenza	147858
724.2 Lumbago	137345
V76.12 Screen mammogram NEC	129445
V70.0 Routine medical exam	127848

Disease	count
530.81 Esophageal reflux	121064
427.31 Atrial fibrillation	113798
729.5 Pain in limb	112449
414.01 Crnry athrscl natve vssl	104478
285.9 Anemia NOS	103351
786.50 Chest pain NOS	91999
599.0 Urin tract infection NOS	87982
V58.69 Long-term use meds NEC	85544
496 Chr airway obstruct NEC	78585
477.9 Allergic rhinitis NOS	77963
414.00 Cor ath unsp vsl ntv/gft	75519

Disease	count
719.47 Joint pain-ankle	28648
300.4 Dysthymic disorder	28530
268.9 Vitamin D deficiency NOS	28455
V72.81 Preop cardiovsclr exam	27897
724.3 Sciatica	27604
787.91 Diarrhea	27424
V2.21 Supervis oth normal preg	27320
365.01 Opn angl brderln lo risk	26033
379.21 Vitreous degeneration	25592
424.1 Aortic valve disorder	25425
616.10 Vaginitis NOS	24736
702.19 Other sborheic keratosis	24453
380.4 Impacted cerumen	24046

Out of 135K patients who had laboratory data

Top lab test results

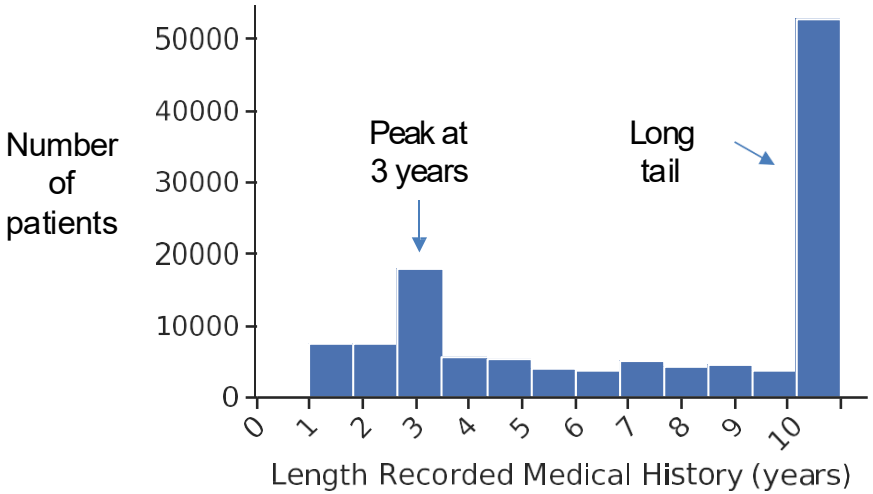
Lab test	
2160-0 Creatinine	1284737
3094-0 Urea nitrogen	1282344
2823-3 Potassium	1280812
2345-7 Glucose	1299897
1742-6 Alanine aminotransferase	1187809
1920-8 Aspartate aminotransferase	1187965
2885-2 Protein	1277338
1751-7 Albumin	1274166
2093-3 Cholesterol	1268269
2571-8 Triglyceride	1257751
13457-7 Cholesterol.in LDL	1241208
17861-6 Calcium	1165370
2951-2 Sodium	1167675

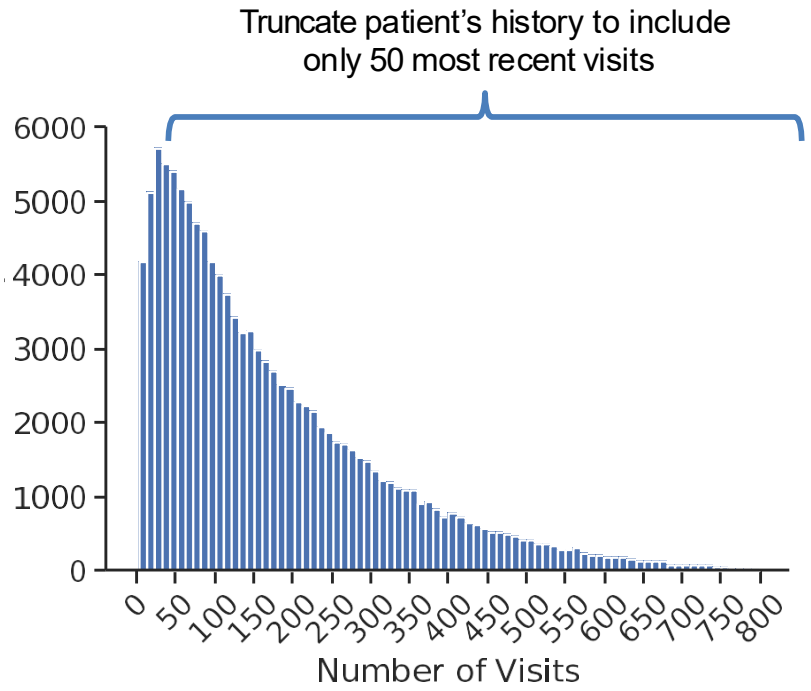
Lab test	
2085-9 Cholesterol.in HDL	1155666
718-7 Hemoglobin	1152726
4544-3 Hematocrit	1147893
9830-1 Cholesterol.total/Cholesterol.in HDL	1037730
33914-3 Glomerular filtration rate/1.73 sq M.predicted	561309
785-6 Erythrocyte mean corpuscular hemoglobin	1070832
6690-2 Leukocytes	1062980
789-8 Erythrocytes	1062445
787-2 Erythrocyte mean corpuscular volume	1063665

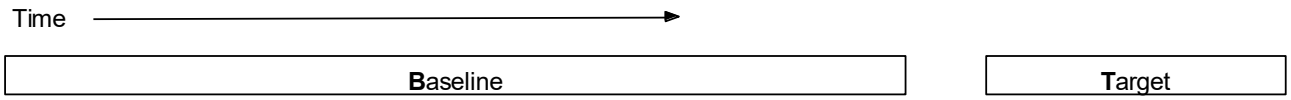
Lab test	
770-8 Neutrophils/100 leukocytes	952089
731-0 Lymphocytes	943918
704-7 Basophils	863448
711-2 Eosinophils	935710
5905-5 Monocytes/100 leukocytes	943764
706-2 Basophils/100 leukocytes	863435
751-8 Neutrophils	943232
742-7 Monocytes	942978
713-8 Eosinophils/100 leukocytes	933929
3016-3 Thyrotropin	891807
4548-4 Hemoglobin A1c/Hemoglobin.total	527062

Count of people who have the test result (ever)

There may be a varying amount of history per patient



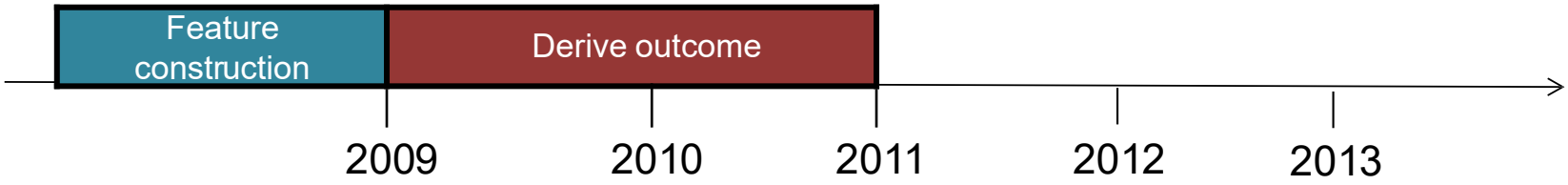




Target = *f*(Baseline)

- How to represent Baseline and Target?
- What class of models do we consider for *f*?

Framing for supervised ML



Exclusion criteria:

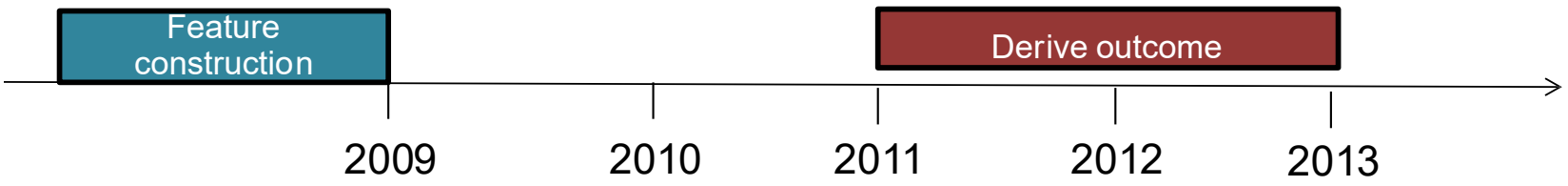
- Diabetes diagnosis (according to our rule) observed prior to January 1, 2009
- Less than 6 months of enrollment in feature construction window
- Member left health insurance prior to Jan. 1, 2011

What if someone is diagnosed with diabetes in 2012?

Why not model as "patient develops diabetes anytime after 2009"?

[Razavian, Blecker, Schmidt, Smith-McLallen, Nigam, Sontag. *Big Data.* '16]

Framing for supervised ML

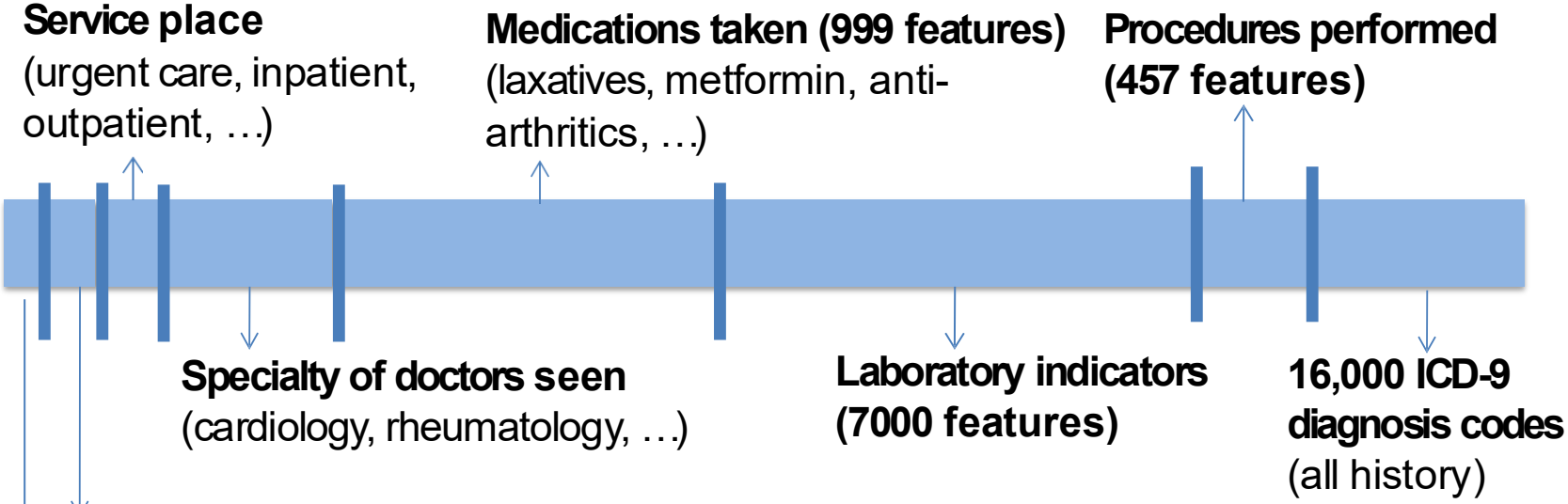


Exclusion criteria:

- Diabetes diagnosis (according to our rule) observed prior to January 1, ~~2009~~ 2011
- Less than 6 months of enrollment in feature construction window
- Member left health insurance prior to Jan. 1, ~~2011~~ 2013

[Razavian, Blecker, Schmidt, Smith-McLallen, Nigam, Sontag. *Big Data.* '16]

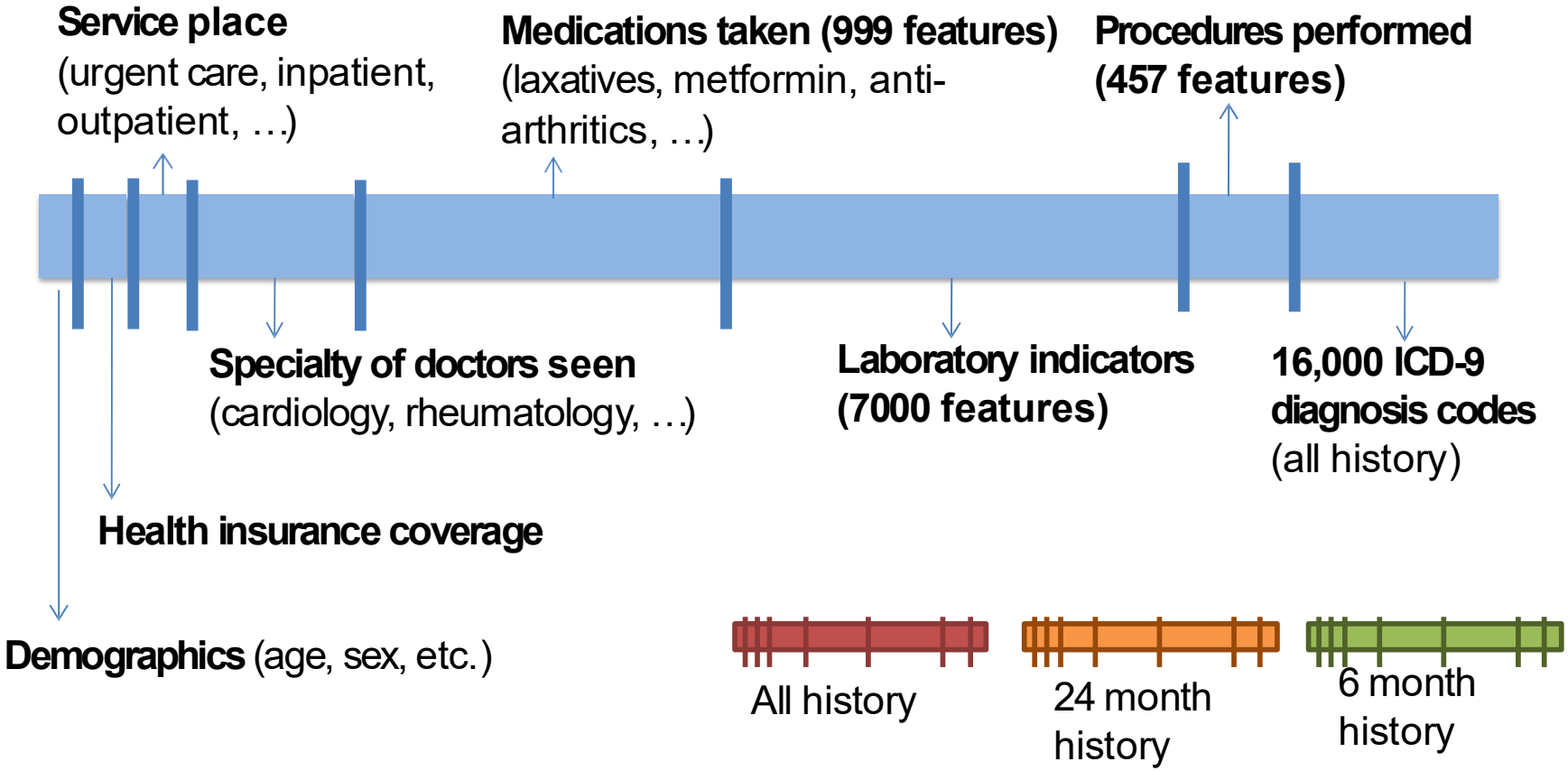
Encoding the longitudinal health data



For the 1000 most frequent lab tests:

- Was the test ever administered?
- Was the result ever low?
- Was the result ever high?
- Was the result ever normal?
- Is the value increasing?
- Is the value decreasing?
- Is the value fluctuating?

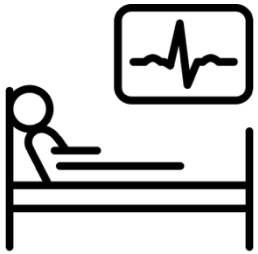
Encoding the longitudinal health data



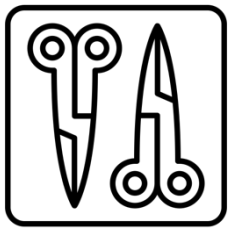
10s-100s of thousands of features

Example

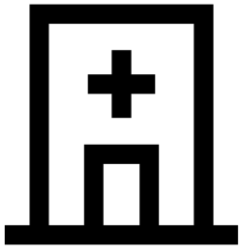
Over a horizon of 3 to 9 months into the future, predict:



End of Life
(EOL)



Surgical
Procedures
(Surgery)



Likelihood of
Hospitalization
(LoH)

Evaluate using de-identified dataset of ~120K Medicare Advantage members

Kodialam et al., *Deep Contextual Clinical Prediction with Reverse Distillation*, AAAI '21

SARD Model

- Visit embeddings summarize patient state at a visit (vs. representing each variable's history separately)
  - 37K codes represented

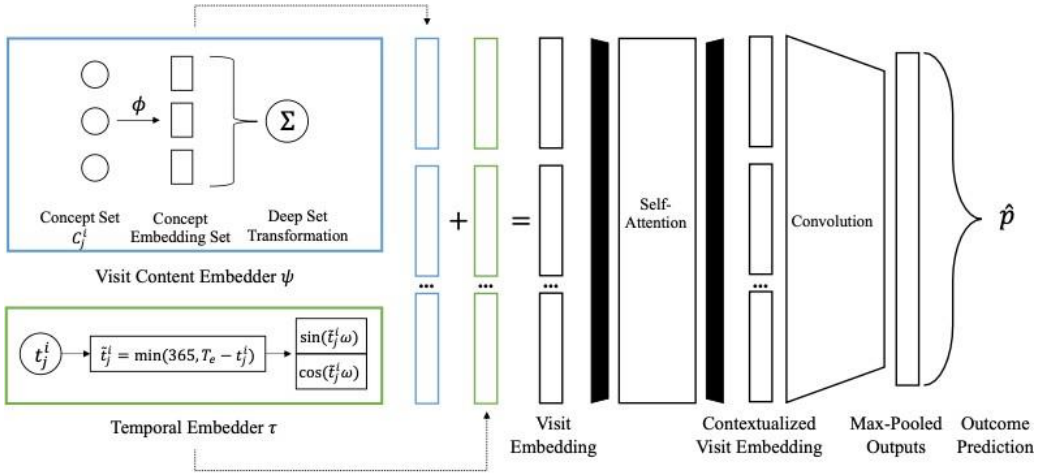


Figure 1: SARD Architecture for Longitudinal Claims Data

ML methods compared

- **SARD** (Kodialam et al. 2021)
- **BEHRT** (Li et al. 2020): another transformer-based neural network for claims data
- **RETAIN** (Choi et al. 2016): a recurrent neural network designed with interpretability in mind
- **Windowed linear model** (Razavian et al. 2015)

Kodialam et al., *Deep Contextual Clinical Prediction with Reverse Distillation*, AAAI '21

Results on the 3 prediction tasks

AUC-ROC scores on test set

Model \ Task Name	EoL	Surgery	LoH
L_1-reg. logistic regression [Razavian et al. 2015]	83.4	79.2	73.1
RETAIN [Choi et al. 2016]	82.2	79.8	72.5
BEHRT [Li et al. 2020]	83.1	80.3	71.2
SARD	**85.6**	**83.1**	**74.3**

Kodialam et al., *Deep Contextual Clinical Prediction with Reverse Distillation*, AAAI '21

Exit pass

https://z.umn.edu/ml4h_exitpass

